

Solutions to Homework #2

Problem 1

(a) We are given the fact that

$$\int_{-\infty}^{\infty} \text{rect}(\lambda) \text{rect}(t - \lambda) d\lambda = \text{tri}(t).$$

This identity holds for all $t \in \mathbb{R}$. Now we consider

$$\begin{aligned} \int_{-\infty}^{\infty} \text{rect}(b\lambda) \text{rect}(b(t - \lambda)) d\lambda &= \int_{-\infty}^{\infty} \text{rect}(b\lambda) \text{rect}(bt - b\lambda) d\lambda \\ &\stackrel{(a)}{=} \int_{-\infty}^{\infty} \text{rect}(s) \text{rect}(bt - s) \frac{ds}{b} \\ &= \frac{1}{b} \underbrace{\int_{-\infty}^{\infty} \text{rect}(s) \text{rect}(bt - s) ds}_{(*)}, \end{aligned}$$

where (a) holds by applying the change of variable $s = b\lambda$, which implies $ds = b d\lambda$. The integral in $(*)$ is interpreted as the evaluation of the convolution of $\text{rect}(t)$ and $\text{rect}(t)$ at bt , which is written as $\text{tri}(bt)$. Therefore, we obtain the desired result.

(b) We use the multiplication property of CTFT, which is given by

$$\mathcal{F}_{\text{CTFT}}\{x(t)y(t)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\theta) Y(j(\Omega - \theta)) d\theta.$$

Since

$$\mathcal{F}_{\text{CTFT}}\{\text{sinc}^2(at)\} = \frac{1}{a} \text{rect}\left(\frac{\Omega}{2\pi a}\right),$$

due to the multiplication property, we have

$$\begin{aligned} \mathcal{F}_{\text{CTFT}}\{\text{sinc}^2(at)\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{a} \text{rect}\left(\frac{\theta}{2\pi a}\right) \frac{1}{a} \text{rect}\left(\frac{\Omega - \theta}{2\pi a}\right) d\theta \\ &= \frac{1}{2\pi a^2} \underbrace{\int_{-\infty}^{\infty} \text{rect}\left(\frac{\theta}{2\pi a}\right) \text{rect}\left(\frac{\Omega - \theta}{2\pi a}\right) d\theta}_{(*)}. \end{aligned}$$

. The integral in $(*)$ is computed as

$$(*) = \text{rect}(bt) * \text{rect}(bt)|_{t=\Omega} = \frac{1}{b} \text{tri}(b\Omega),$$

where $b = \frac{1}{2\pi a}$. By plugging in this back, we obtain

$$\mathcal{F}_{\text{CTFT}}\{\text{sinc}^2(at)\} = \frac{1}{a} \text{tri}\left(\frac{\Omega}{2\pi a}\right).$$

(c) By Euler's identity, we have

$$\cos(10\pi at) = \frac{e^{j10\pi at} + e^{-j10\pi at}}{2}.$$

Then we use the modulation (freq-shift) property, which says

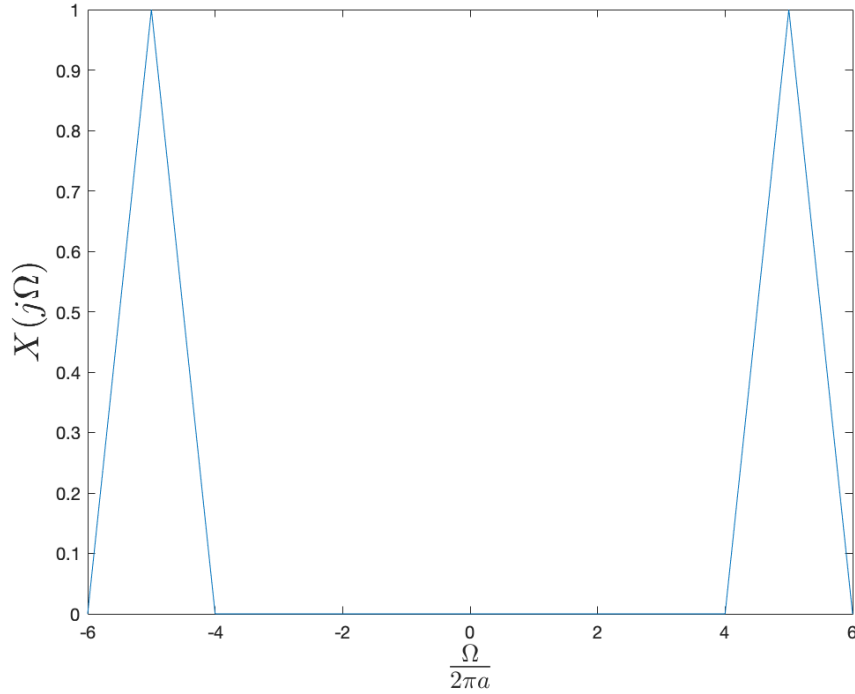
$$\mathcal{F}_{\text{CTFT}}\{e^{j\theta t}x(t)\} = X(j(\Omega - \theta)).$$

Therefore, from the result of part (b), we obtain

$$\mathcal{F}_{\text{CTFT}}\{\cos(10\pi at)\text{sinc}^2(at)\} = \frac{1}{2a}\text{tri}\left(\frac{\Omega - 10\pi a}{2\pi a}\right) + \frac{1}{2a}\text{tri}\left(\frac{\Omega + 10\pi a}{2\pi a}\right).$$

(d) The figure can be generated by the following Matlab script.

```
t = -10:0.001:10;
tri = @(t) max(1-abs(t),0);
f = @(t) max(tri(t-5),tri(t+5));
xlabel(' \frac{\Omega}{2\pi a} ', 'interpreter', 'latex', 'FontSize', 20);
ylabel(' X(j\Omega) ', 'interpreter', 'latex', 'FontSize', 20);
```



Problem 2

Let $X_c(j\Omega)$ and $Y_c(j\Omega)$ denote the Fourier transform of $x(t)$ and $y(t)$, respectively. By the conjugation and time-reversal properties of the Fourier transform, the transform of $w(t) = x^*(t)$ is given by $W_c(j\Omega) = X_c^*(-j\Omega)$. Define $z(t) = x^*(t)y(t) = w(t)y(t)$. Then the Fourier transform $Z_c(j\Omega)$ of $z(t)$ can be written as

$$Z_c(j\Omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} W_c(j(\Omega - \lambda))Y_c(j\lambda)d\lambda = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c^*(j(\lambda - \Omega))Y_c(j\lambda)d\lambda.$$

Evaluating $Z_c(j\Omega)$ at $\Omega = 0$, we obtain

$$Z_c(j0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c^*(j(\lambda))Y_c(j\lambda)d\lambda.$$

On the other hand, by the definition of the Fourier transform, $Z_c(j0)$ is also written as

$$Z_c(j0) = \int_{-\infty}^{\infty} z(t)dt = \int_{-\infty}^{\infty} x^*(t)y(t)dt.$$

Problem 3

(a) Note that $x(t)$ is written as

$$x(t) = \gamma_1\delta(t - \tau_1) + \gamma_2\delta(t - \tau_2) + \gamma_3\delta(t - \tau_3), \quad 0 \leq t < 2\pi,$$

and $x(t + 2\pi) = x(t)$. The summation over shift indexes l is not necessary within the interval $[0, 2\pi)$. The Fourier series coefficient is calculated by the analysis equation by using the sifting property of Dirac's delta:

$$\begin{aligned} a_k &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{m=1}^3 \gamma_m \delta(t - \tau_m) e^{-jkt} dt \\ &= \frac{1}{2\pi} \sum_{m=1}^3 \gamma_m \int_0^{2\pi} \delta(t - \tau_m) e^{-j\tau_m k} dt \\ &= \frac{1}{2\pi} \sum_{m=1}^3 \gamma_m e^{-j\tau_m k}. \end{aligned}$$

(b) Expanding $y(t)$, we obtain

$$y(t) = b_3 e^{j3t} + b_2 e^{j2t} + b_1 e^{jt} + b_0,$$

where

$$\begin{aligned} b_3 &= 1 \\ b_2 &= -e^{j\tau_1} - e^{j\tau_2} - e^{j\tau_3} \\ b_1 &= e^{j\tau_1}e^{j\tau_2} + e^{j\tau_1}e^{j\tau_3} + e^{j\tau_2}e^{j\tau_3} \\ b_0 &= -e^{j\tau_1}e^{j\tau_2}e^{j\tau_3}. \end{aligned}$$

This expression is already in Fourier-series form, with the Fourier coefficients $\{b_k\}$ determined by the above relations. All other coefficients vanish, i.e., $b_k = 0$ if $k \notin \{0, 1, 2, 3\}$.

- (f) Note that $x(t) = 0$ for all $t \in [0, 2\pi)$ except at $t = \tau_1, \tau_2, \tau_3$. Meanwhile, $y(t) = 0$ exactly at those same points. Hence, we have

$$x(t)y(t) = 0, \quad \forall t \in [0, 2\pi).$$

Since both $x(t)$ and $y(t)$ are 2π -periodic, their product $x(t)y(t)$ is also 2π -periodic. Consequently, $x(t)y(t) = 0$ for all $t \in \mathbb{R}$. Then, by the multiplication property of the Fourier series, the fact $x(t)y(t) = 0$ implies that their corresponding Fourier-series coefficient sequences $\{a_k\}$ and $\{b_k\}$ satisfy

$$\{a_k\} * \{b_k\} = 0$$

or equivalently

$$\sum_{m=-\infty}^{\infty} a_{k-m}b_m = 0, \quad \forall k \in \mathbb{Z}. \quad (1)$$

Since $\{b_k\}$ is supported on $\{0, 1, 2, 3\}$, and $b_3 = 1$, Equation (1) reduces to

$$\begin{bmatrix} a_k & a_{k-1} & a_{k-2} \end{bmatrix} \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} = -a_{k-3}. \quad (2)$$

- (d) By collecting Equation (1) for $k = 3, 4, 5$ we obtain

$$\begin{bmatrix} a_3 & a_2 & a_1 \\ a_4 & a_3 & a_2 \\ a_5 & a_4 & a_3 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = - \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}.$$

By the construction of $y(t)$, $e^{j\tau_1}, e^{j\tau_2}, e^{j\tau_3}$, τ_1, τ_2, τ_3 are the roots of $z^3 + b_2z^2 + b_1z + b_0 = 0$. We have shown that the parameters τ_1, τ_2 , and τ_3 can be computed from $a_0, a_1, \dots, a_5$.